



Time = 20 Minutes

Quiz No 03

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [20 points] [Counting]

CLO Assessed – CLO 03, Apply the axioms of probability and compute probabilities by counting. (Cog - 3)

We roll a 6-sided die with numbers 1 to 6 each having probability p_i (where i is 1, 2, ..., 6). If we roll the die 6 times, answer the following questions: -

Part a [2 points] Find the total numbers of possible ordered outcomes (permutations).

Part b [2 points] In how many ways can we get an odd number (1 or 3 or 5) on rolls 1, 3 and 5 and an even number (2 or 4 or 6) on rolls 2, 4 and 6?

Part c [2 points] In part (a), how many of those outcomes have all the six numbers (1 to 6) appearing?

Part d [2 points] Find the probability that each consecutive roll results in a bigger number than the previous roll.

Part e [3 points] Find the probability of a **particular sequence** where we get an odd number (1 or 3 or 5) on rolls 1, 3 and 5 and an even number (2 or 4 or 6) on rolls 2, 4 and 6?

Part f [3 points] Find the probability of outcomes where all the six numbers appear (see part c).

Part g [6 points] We define the following two events: -

1. **Event A** - we get number 6 four times in the six rolls.
2. **Event B** - the first and last rolls are number 6 in the six rolls.

Calculate the Probability, $P(B|A)$. Note – the answer needs to be completely simplified in its final form.

(a) $6 \times 6 \times 6 \times 6 \times 6 \times 6 = 6^6$ — 2 pt

(b)

3	3	3	3	3	3
1	2	3	4	5	6

 $= 3^6$ — 2 pt

(c)

6	5	4	3	2	1
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 $= 6!$ — 2 pt

(d) There is only 1 Sequence of Roll that Satisfies the Condition
No 1, 2, 3, 4, 5, 6 = $P_1 \cdot P_2 \cdot P_3 \cdot P_4 \cdot P_5 \cdot P_6$
 $= P_1 \cdot P_2 \cdot P_3 \cdot P_4 \cdot P_5 \cdot P_6$ — 2 pt

(e) we can get 1, 3 or 5 on roll No 1, 3 & 5 and 2, 4 or 6 on roll No 2, 4 & 6. 1 pt

$$\Rightarrow \text{Probability} = \left(P_1 + P_3 + P_5 \right) \left(P_2 + P_4 + P_6 \right)$$

of
Particular Sequences.

2 pt

(f) $P[\text{outcomes where all 6 Numbers appear}]$

$$P[\text{all 6 N\acute{o}s}] = P_1 \cdot P_2 \cdot P_3 \cdot P_4 \cdot P_5 \cdot P_6 \quad \text{--- (A)}$$

a particular
Sequence

1 pt

No. of ways this can appear each with
Probability = (A)

1 pt

$$P[\text{outcomes w/ all 6 N\acute{o}s}] = 6! \cdot P_1 \cdot P_2 \cdot P_3 \cdot P_4 \cdot P_5 \cdot P_6$$

1 pt

(g) $P(B|A) = \frac{P(B \cap A)}{P(A)}$

Given A, all possible sequences with 4 times 6's ~~are~~ have same probability. Therefore, we can invoke Counting.

No. of Sequences in Event A = $6C_4$

Since $B \cap A$ also have 4, 6's, therefore same Prob.

$B \cap A$ have total possible combinations = $4C_2$

1 pt

$$P(B|A) = \frac{4C_2}{6C_4} = \frac{6}{15}$$

3 pt

2 pt